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RESEARCH ARTICLE

Unequal Content: Search Engine Personalization Influences Exposure to Information Regarding the Israel–Palestine Conflict

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ABSTRACT Search engines serve as primary gateways to information, and users tend to trust them, despite being opaque algorithms and customizing results in non-transparent ways. Personalization, such as tailoring content based on a user's location, is useful for routine content searches (e.g., finding nearby restaurants). However, this same personalization becomes problematic – and potentially harmful – when users seek information on sensitive or contentious geopolitical topics. In this work, we present the first large-scale measurement of how search engine curation and personalization affect users across different countries with distinct regional, political, and cultural traits. We developed a novel, privacy-preserving, bot-based methodology that simulates online user profiles embedded with cumulative regional traits. These customized bots were deployed from different locations and conducted identical searches, either on the Israel-Palestine conflict (e.g. ''Hamas'') or neutral queries (e.g. ''dinner recipe''), using both Google and DuckDuckGo. Our study addresses two main questions: 1) How do search results for the same query differ across countries, and which factors drive these differences?; and 2) Does personalization vary more significantly for conflict-related or for general queries? Our findings show that search result rankings are significantly influenced by location, interface language, and simulated browsing context, with the strongest effects observed for conflict-related queries. In many cases, these differences led to non-neutral portrayals of the same event, depending on the user's presumed region. These results highlight how search engine personalization can shape public understanding of global events, creating fragmented or biased informational environments. This underscores the importance of educating users about the invisible filters that influence their access to information and, from a policy standpoint, these findings reveal a pressing need for greater transparency and accountability in how search engines personalize politically sensitive content. This is especially urgent given their outsized role in shaping democratic discourse and public opinion worldwide.

INDEX TERMS Search engine audit, filter bubble effect, personalization, algorithmic transparency.

I. INTRODUCTION

Access to high-quality information is a central tenet of democratic regimes, allowing citizens to make informed decisions. In recent decades, the way people seek information has shifted from traditional to online media [1], [2], [3], with search engines playing a crucial role as gatekeepers. Google, for example, processes around 8.5 billion daily queries [4], [5], highlighting its influence on global information consumption.

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While this shift has greatly enhanced the speed and convenience of accessing information, it also raises concerns about how search results shape users' worldviews. Studies show that people searching for news place high trust in search engine results [6], [7], [8], [9], [10], often seeing these results as the most relevant [11]. Simultaneously, search engines explicitly acknowledge their use of personalization algorithms that tailor content based on user-specific factors, such as location, language, preferences, and search history [12], [13], [14], [15]. While such tailoring can be very useful for routine tasks, such as looking for a service, it becomes problematic for sensitive topics, such as political news, where customized results can introduce bias and reinforce ''filter bubbles'' [15], [16]. Given the opaque nature of search engine algorithms, which tend to monetize clicks, it is difficult to determine *how much of this tailoring affects controversial topics compared to less sensitive ones*. If differences exist, greater transparency is needed to explain how and why search engines suggest user-specific results.

Prior research has largely focused on either non-sensitive topics [15], [17] or political events – almost exclusively elections [18], [19], [20], [21], [22], [23], [24]. Moreover, these works often focused on isolated factors, typically analyzing a single feature at a time (e.g. location [14], [15]), with an emphasis on URL differences rather than a comprehensive assessment of content type or slant [15], [20], [22], [23]. This narrow approach leaves a gap in the full understanding of the scope, extent and implications of search engine personalization. In this work, we address these limitations by presenting, to the best of our knowledge, the first empirical study of search engine personalization to:

(i) **Explore country-level location-based personalization on geopolitical topics:** while previous studies have documented the impact of location on search engine results, little is known about how users from different countries receive search results for identical queries, particularly for topics in which geographic context is central to interpretation.

(ii) **Investigate the combined effect of multiple regional traits:** in contrast to previous studies that isolate the effects of single features, we propose a method to evaluate how a combination of user online attributes (location, browser language, and local browsing history) may interact to influence search engine personalization.

(iii) **Focus on a globally polarizing case study:** unlike studies centered on broad or domestic political topics, we analyze the Israeli-Palestinian conflict, a setting where location fundamentally shapes interpretation. Characterized by divergent political, cultural, and linguistic narratives, this conflict serves as an ideal lens to examine how search personalization (driven by location and other identity markers) can amplify politically consequential biases.

To this end, we developed a methodology using web crawlers (bots) designed to closely mimic human browsing behavior. These bots, customizable by location (via proxies), language, and user profiles (via cookies), were programmed to perform queries at a human-like pace, scroll through results in a way that closely resembled real users – robust enough to evade detection mechanisms. Specifically, we queried two search engines – Google, the most widely used globally, and DuckDuckGo, a privacy-oriented alternative – on conflict-specific (e.g.: ''Hamas attacks'') and general topics (e.g: ''Popular books 2024''). Our analysis focuses on three main research questions: (1) Does personalization affect polarizing and non-polarizing queries differently, and is it amplified by additional user identity traits (e.g., language, browsing history)? (2) Do variations in URLs reflect fundamentally different content recommendations? (3) Do these differences lead to distinct representations of the Israeli–Palestinian conflict depending on the user's location?

Performing this study presented several challenges. First, search engines are proprietary and opaque, making it difficult to systematically examine how their algorithms work. Second, it is essential to mimic realistic user behavior across different regions, languages, and browsing habits without introducing bias. Third, it is not trivial to distinguish between neutral and politically charged queries to draw meaningful conclusions, and define reliable metrics to measure the variations in search engine results. This paper addresses these challenges and makes the following key contributions:

1. Crawling method with incremental profiling: Our bots were developed to increasingly resemble users from different countries (Israel, Saudi Arabia, Brazil, United States) with different perspectives and interests regarding the Israel-Palestine conflict. *Type 1* bots differed only by location. *Type 2* bots included browser language customization. *Type 3* bots added browsing history through exposure to regional news content. This staged profiling allowed us to assess both the level of personalization and the specific contribution of each attribute.

2. Metrics for measuring customization: We analyzed both the degree and nature of variation in search results using a metricbased on Rank-Biased Overlap (RBO) [25], which gives more weight to top-ranking results. We also applied qualitative methods to assess content differences and potential ideological leanings. This dual approach enables a broader application, beyond this specific case study.

3. Significant findings: Our study reveals stronger levels of search engine personalization than previously reported, particularly for politically sensitive queries [18], [20], [26]. The results indicate that personalization is significantly more pronounced in conflict-related searches than in general queries, often leading to divergent portrayals of the Israeli–Palestinian conflict. Moreover, despite claims by search engines that they limit personalization, browsing history had a measurable effect on results variation, including for DuckDuckGo.

II. BACKGROUND AND RELATED WORK

Algorithmic personalization, particularly recommendation bias, has been extensively studied in the context of social media feeds [27], [28], [29] and content platforms [30], [31], [32]. Studies of Facebook, TikTok, and Twitter/X have

investigated how tailoring of content reinforces existing preferences [33], contributing to phenomena such as *filter bubbles* [34] and/or *echo chambers* [33] and polarization [35]. Similar concerns have been raised in platforms such as YouTube [30], [31] and Amazon [36], where recommendation algorithms influence both consumer behavior and public opinion. While there is disagreement on the extent and pervasiveness of these phenomena [29], [33], [34], a large body of work highlights risks of polarization, reinforcement of biases, and reduced exposure to diverse viewpoints. This has led to significant discussions on how recommender systems might be redesigned to promote more democratic and less biased information ecosystems [37], [38].

By contrast, and despite occupying a key position as gatekeepers to information, search engines have received comparatively less attention. First, they are often the starting point for active information seeking – users repeatedly turn to search engines to ''find the truth'' or to obtain knowledge [39], [40]. Second, factors such as keyword matching, site authority, user location, and language contribute to how results are selected and ranked [41], [42], but the exact mechanisms remain opaque. Third, they are highly trusted and the trust placed in their results can be so significant that even subtle ranking manipulations have been shown to shape users' attitudes and decisions [43].

Still, the literature on search engine personalization remains divided on two key questions: (1) the extent of curation present in search results, and (2) the magnitude of its impact on users, including whether search engines generate filter-bubble-like effects.

In this work, we examine how both well-documented features (e.g., location, as acknowledged by search engines and confirmed in prior studies) and less explored factors influence the order, content, and leaning of search engine results, particularly in the context of geographically and politically sensitive topics such as the deeply polarized Israeli-Palestinian conflict. To better situate this study, the remainder of this section first outlines the public policies regarding personalization of search engine results, and then reviews prior audit studies of search engine algorithms.

A. SEARCH ENGINE POLICIES

Search engines typically outline their public policies regarding personalization and customization of results on their websites. To understand what kinds of behavior can be expected under these policies, we describe the approaches of Google and DuckDuckGo below and summarize their main claims in Table 1. Although search engines often distinguish between more general factors used to curate results – customization – and more personal cues, such as browsing history, that tailor results to an individual user – personalization – throughout this paper we use personalization to refer to any form of tailoring. This simplification follows prior literature and reflects the view that any influence on search results can shape users' perspectives on particular topics.

TABLE 1. Customization features disclosed by search engines' terms of service. ✓ - feature impacts search results ✗ - feature does not impact search results ● - feature impacts results when user is logged in ❍ - policy is unclear on whether feature impacts results.

✓ - feature impacts search results
✗ - feature does not impact search results
● - feature impacts results when user is logged in
❍ - policy is unclear on whether feature impacts results

Feature	Google	DuckDuckGo
Location	✓	~proxy
Language	✓	✓
Device Type	✓	❍
Search History (within platform)	●	✗
Browsing History	✗	✗
Previous Content Engagement	●	✗
Private Data (e.g., Email)	✗	✗

Google is the most widely used search engine, and claims that results may differ from user to user based on ''time, context and personalization'' [44]. Time differences arise from data being updated at various speeds across data centers. Context includes aspects such as location, language, device type (screen, mobile or desktop), and related results, which may appear if a user engages with specific content on the results page. Personalized results – specifically tailored to an individual user profile – are claimed to be tied to a Google Account, only occurring when users are logged in.

DuckDuckGo is known for its privacy-concerned model, explicitly stating that it does not log IP addresses, browsing history, or search history to determine results. However, while it does not use exact user locations, it provides geographically tailored results through proxies [12].

B. SEARCH ENGINE AUDITS

Given the proprietary nature of search engine algorithms, researchers have relied on external audits to evaluate the extent and possible negative consequences of search engine personalization. In 2013, Hannak et al. [15] introduced a methodology based on comparing the results of a ''real'' person with those of a ''fresh'' profile for the same query, through a browser extension. They showed that ''location'' and ''being logged-in'' were the most important factors to induce differences in search results. Since this pioneering work, other studies have relied solely on data donation, both from general users [17], [20], [45], [46] and from search engine companies themselves [47]. More recently, research has shifted towards agent-based methodologies, closer to the one proposed and developed in this work [24], [48], [49]. However, our approach distinguishes itself through the incremental development, robustness, and detailed configuration of our agent infrastructure – here referred to as bots.

Notably, due to its importance, most previous studies have focused on political customization in the context of

electoral processes, and almost solely audit Google [18], [19], [20], [22], [24], [50]. These often examine the effects of different factors in isolation (typically either location [14], [17] or browsing history [24]), and have produced mixed findings: while some earlier research found no significant personalization in search results based on users' political profiles [18], [20], [26], more recent studies suggest that a user's political leaning can influence the political bias of results, especially for top news listings [22], [24]. Moreover, location has been shown to have a stronger influence on local queries than on political ones [14], [17]. Methodologically, this literature primarily compares URLs, only rarely investigating differences in the content that these URLs direct to. Important exceptions are the work of Le et al. [24] – measuring the leaning of search engine results' domains, but not of individual URLs –, of Robertson et al. [22] – grouping search results in categories –, and of Hu et al. [21] – scoring partisan bias of results' snippets and their corresponding web pages.

More recently, scholars have also turned their attention to the inherent biases of search engines themselves [51], beyond those possibly introduced by personalization. This body of work has documented systematic disparities, including gender [52], [53], race [53], [54], and source-related biases [55], [56], [57], as well as the amplification of inaccurate or conspiracy-promoting content [58], [59]. In parallel, another line of research has shown that biases can also arise from users themselves, particularly through the ways in which queries are framed and selected, which significantly shape the results returned [46], [60]. These findings raise the question of whether similar structural and self imposed biases might emerge when users seek information on politically polarizing topics – where the stakes of representation and accuracy are particularly high.

Therefore, our work aims to bridge these perspectives by exploring not only the impact of individual features on a single search engine, but the more natural setting of a combination of several significant factors – those highlighted by previous studies and publicly acknowledged by search engines (Table 1). Our encompassing methodology analyzes how these factors increasingly influence search outcomes in terms of URLs, content, and leaning. Specifically, we configure bots to simulate users from different regions, each carrying multiple identity signals (e.g., language settings, location, and browsing history), and compare the results when performing two types of queries: (1) popular and general queries, and (2) queries on topics where location is a critical factor.

III. METHODOLOGY

Our methodology relies on a web crawling system, specifically developed to explore how search engine results may change for different user profiles. It enables the execution of web crawlers (*bots*) that can mimic typical online user behavior, be deployed to different locations, and make queries that are either related to the Israeli-Palestinian conflict or more general, non-sensitive queries. Figure 1 summarizes our methodology for developing and implementing this querying and auditing system. It is divided into three main stages: *1)* design of each bot, including its incremental feature design and deployment, *2)* audit of search engine results, and *3)* evaluation of search result differences in terms of URLs, content, and possible leaning.

A. BOTS AND THEIR INCREMENTAL DESIGN

Using bots over real user data offers several advantages: (1) precise control over key profile features to isolate their impact on search results; (2) it is an ethical and privacy protecting approach, avoiding sensitive user data collection; and (3) does not require asking users to perform sensitive searches, which could influence their future experiences [61].

1) BOT CONSTRUCTION AND OPTIMIZATION

We developed each bot by extending the functionalities of OpenWPM[1] (Version 0.28.0), a community-trusted and flexible instrument for internet measurements and automating web browsing [62]. The bots were built using Selenium and deployed using version 123 of the Firefox browser, which can be parallelized and allows control of location, through proxy configuration, and browser settings such as language, cookies, and other forms of tracking, as detailed below. To minimize bot detection by search engines, we prioritized fresh browser instances, used residential proxies, and the bots were programmed to mimic human-like behavior, scrolling through pages, clicking on page elements, and typing queries at a human-like pace. Together, these were sufficient to bypass detection and allowed the completion of nearly all queries (Appendix, Figure A4): despite the technical challenges posed by search engine defenses, the process was robust, yielding a substantial dataset for analysis (7,496 first-page search results).

Location: To change the location of the bots, Firefox settings were adjusted to use residential ISP proxies provided by Bright Data.[2]

ISP proxies act as intermediate remote services for Internet access, masking the original source of the HTTP request. Using this system, the bots were placed in one of four locations: Israel (IL), Saudi Arabia (SA), Brazil (BR), and New York, United States (US-NY). Israel was selected due to its direct involvement in the conflict, with a majority Jewish and Hebrew-speaking population. Since Bright Data did not offer ISP proxies for Palestine, Lebanon, or Iran, Saudi Arabia was chosen for its majority Muslim and Arabic-speaking population. New York and Brazil were chosen as not-directly involved controls: New York due to its sizable Jewish community [63], and the U.S.'s strong support for Israel, and Brazil as a more neutral location [64], with less support for Israel, and having a predominantly Christian population.

[1] https://github.com/openwpm/OpenWPM
[2] https://brightdata.com

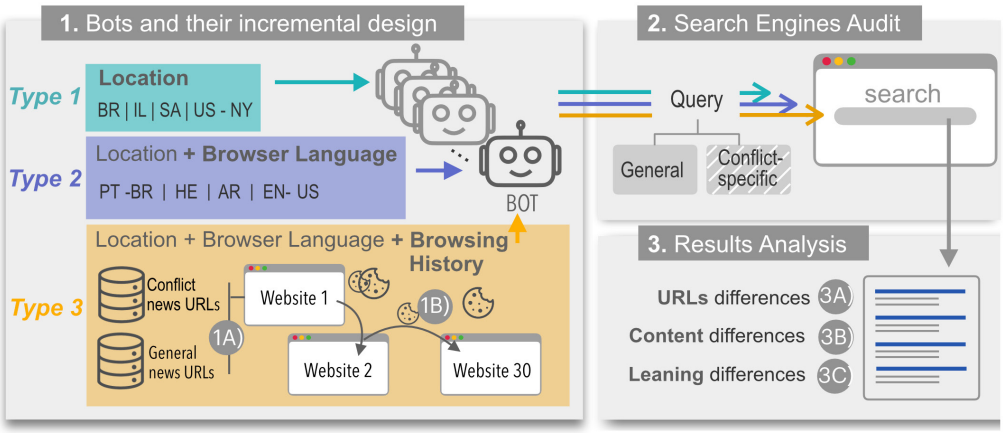


FIGURE 1. ***Methodological scheme.*** **Three types of bots were deployed (1): *Type 1* (green) only differ in location, *Type 2* (violet) in location and language, *Type 3* (yellow) in location, language, and browsing history, having collected cookies beforehand. These bots are directed to search engines and make "general" or "conflict specific" queries (2). The results are analyzed according to URL, content type, and leaning (3).**

Browser Language: To simulate Internet access with browsers in the predominant language of the aforementioned locations, Firefox language settings for each bot were adjusted accordingly: Hebrew (he) for Israel, Arabic (ar) for Saudi Arabia, Portuguese (pt) for Brazil, and English (en) for the United States.

Browsing History: Online activity and browsing history (website visits, content interactions, etc.) can be tracked through cookies placed by first and third party sites, or by the specific configuration of the device and browser used to access the internet (fingerprinting) [65], [66], [67]. By instructing the bots to visit different online content, and saving the resulting tracking data, we aimed to create distinct browsing histories outside of the search engine platforms (*Type 3* bots).

Site selection: Two URL lists were created for each of the bot locations: 1- Conflict news: local news that contained one of the keywords "Palestine',' "Israel", "Hamas", and "Netanyahu"; 2- General news: local news that included the keywords "movie", "health", "well-being", "dinner recipe", and "sports". When necessary, the keywords were translated from English into the three other languages, to collect local news in the most spoken language of the country. For all lists (one conflict-related news list and one general news list for each of the four locations), URLs were obtained from October 7, 2023 to March 30, 2024, through Media Cloud,[3] a free and community supported tool that allows retrieval of news articles by keywords. To minimize selection bias and to ensure that only local news sources were included for each country, we applied consistent rules across all cases: (1) use of Media Cloud's national collections [68], (2) reliance on the same keywords to identify relevant news within those collections, which together yielded a total of 230,598 URLs, and (3) random sampling of the same number of URLs (20) per location. The final list of visited domains is provided in Table A11 in Appendix.

To generate browsing profiles for *Type 3* bots, newly created bots, with no history or cookies, were instructed to visit the random sample of 20 websites sequentially for each of the two lists curated for that particular location (Figure 1, 1A). During these visits, the bots collected and saved all tracking data, which included cookies placed by first and third parties (OpenWPM setting: `tp_cookies = always`), JavaScript fingerprinting files (OpenWPM setting: `js_instrument = True`), and HTTP requests (i.e., calls from external parties), which are collected by default with OpenWPM crawlers – Figure 1, 1B). This data was then stored in Firefox's profiles, which contained the bots' full browsing history and tracking information.

2) INCREMENTAL DESIGN AND DEPLOYMENT

To study the impact of these three features (location, browser language, and browsing history) on search engine results, we deployed three types of bots with an increasing number of features. *Type 1* are bots specified to be solely associated to one of the four locations, *Type 2* are bots that have both different location and browsing languages, and *Type 3* are bots that diverge in their location, browser language, and browsing history (Figure 1). Importantly, the bots of *Type 3* were deployed simultaneously with bots having no browsing history (equivalent to a *Type 2* bot) as a temporal control (Stateless bots). Table 2 presents a summary of the features used by each type of bot, as well as the number (last column) deployed by type.

B. SEARCH ENGINE AUDIT

In our study, two search engines were audited – Google and DuckDuckGo. Google was chosen for being the most used search engine worldwide [69], and DuckDuckGo for its emphasis on privacy [12].

[3]https://www.mediacloud.org/

TABLE 2. Summary of bot types, their features, and respective number of bots associated with each configuration. In the case of Type 3 bots (with cookies) two different browsing histories were used: (Conflict news) if the bot visited news articles related with the conflict and (General news) if the bots has a browsing history of news articles not related with the conflict. (Stateless) corresponds to bots that were released with Type 3 bots but had no browsing history.

	Location	Browser Language	Browsing History	N Bots
Type 1	IL	en-US	-	10
	BR	en-US	-	10
	SA	en-US	-	10
	US-NY	en-US	-	10
Type 2	IL	he	-	10
	BR	pt	-	10
	SA	ar	-	10
	US-NY	en-US	-	10
Type 3	IL	he	Conflict News	3
			General News	3
			Stateless	2
	BR	pt	Conflict News	3
			General News	3
			Stateless	2
	SA	ar	Conflict News	3
			General News	3
			Stateless	2
	US-NY	en-US	Conflict News	3
			General News	3
			Stateless	2

1) AUDIT

For each search engine audit, multiple bots associated with different online user characteristics (see Table 2) were simultaneously deployed. These bots input identical queries and collect the resulting URLs along with their rankings. This process is repeated across various queries and applied to the two search engines. To minimize the risk of carry-over effects (where previous searches could influence current results [15]), each audit began with the same initial user profiles, launched in fresh browser instances.

Three audits were conducted between 30 March and 4 May 2024 (before the inclusion of AI-generated responses and Chatbots in search responses), with each audit lasting an average of two days and involving the deployment of bots of all types (Appendix I, Figure A1). For audits using *Type 2* (location + browser language) and *Type 3* (location + browser language + browsing history) bots, where specifications became more detailed, we configured the bots so that the settings of each search engine were directly comparable. Specifically, we ensured that engines were configured to use the primary language of the country and to prioritize local results. By default, DuckDuckGo uses the browser language, while Google automatically uses the regional "current location" to prioritize local content. To make the settings consistent, we explicitly set both search engines to use local results and the primary language of the country. For reference, see Appendix, Figures A2 (Google) and A3 (DuckDuckGo). Each of these bot configurations was validated by collecting the HTML element that specifies the location and language settings (e.g., "il-HE" for Israel with Hebrew).

To account for potential temporal fluctuations in ranking algorithms, two strategies were employed: (1) conducting simultaneous searches on the same search engine to minimize short-term variations, and (2) performing audits at least twice over distinct time points to address longer-term changes.

All audits were carried out in a local server with two octacore processors and 64 GB of RAM, by setting up a Docker container with Firefox and OpenWPM pre-configured. Within each Docker container, Firefox instances were configured to connect to the Internet through a Bright Data proxy, using OpenWPM's flexible configuration features. The code for our audits methodology can be found at this public repository: 10.5281/zenodo.17642372.

2) SEARCH ENGINE QUERIES

Queries were split into two categories: "General" (denoted by the solid bars in Figures 2 and 3) and "Conflict-Specific" (dashed bars in said figures). The first category comprises 20 popular (and not conflict related) queries such as "Home workout routines", "Popular books 2024", "how to tie a tie" (Table A1, column on the left). The second category, specific to the conflict (e.g., "military complex Al-Shifa hospital", "Hamas rapes"), was defined in collaboration with the NGO HoneyComb, an investigative journalism association that tracks controversial online content, including disinformation. HoneyComb provided 10 distinct topics that were expanded using ChatGPT 3.5, to generate 27 specific queries, which were then reduced to 20 to prevent redundancy (Table A1, column on the right).

A total of 20 queries per group (general and conflict-specific) were executed during the search engine audits for both *Type 1* and *Type 2* bots. As additional complexity exponentially increases the number of combinations, the audit deploying *Type 3* bots performed a subset of 8 queries from the original set (Appendix A, Table A1, gray background). To maintain statistical integrity, each result comparison ensured a consistent number of bots (accounting for occasional IP failures, Appendix A, Figure A5) and queries per location (Appendix A, Figure A4).

C. QUANTIFYING DIFFERENCES IN SEARCH RESULTS

After collecting search engine results and their respective rankings, the differences across distinct bot profiles were analyzed. This analysis was conducted in three stages:

1) QUANTIFYING DIFFERENCES IN URLS

When comparing two lists of URLs returned by search engine queries (i.e., [website_1.com, website_2.com, ...] vs. [website_3.com, website_1.com, ...]), both the links themselves and their order were considered, as users typically prioritize top results over lower-ranked ones [70]. The Rank-Biased Overlap (RBO) metric [71] was used, which calculates how similar two lists are (ranging from 0 for totally

different lists to 1 for equal lists), considering the order of the items and different list sizes [71]:

$$\mathrm{RBO}(S, T, p) = (1-p)\sum_{d=1}^{\infty} p^{d-1} \cdot \frac{X_d}{d} \tag{1}$$

where S and T are two ranked lists, d is the rank, $X_d = |S_{1:d} \cap T_{1:d}|$ and p determines the top-weightiness of the metric. For our analysis, p was set to 0.7, meaning that the first 3 results account for $\approx$ 66% of the total value of the calculated RBO. The differences between two lists of search engine URL results were calculated as:

$$D = 1 - \mathrm{RBO} \tag{2}$$

Specifically, we computed dissimilarity D for pairs of bots with the same location and specifications, as well as for pairs with different locations and specifications (Table 2), while controlling for the number of comparisons when calculating the averages of each condition. Other metrics, such as edit distance [72] and intersections of lists were also applied and yielded similar results.

2) QUANTIFYING DIFFERENCES IN CONTENT

To account for the possibility that the bots received similar results through different local or language specific URLs (e.g. https://en.wikipedia.org/wiki/Palestine vs. https://pt.wikipedia.org/wiki/Estado_da_Palestina), we classified all domains retrieved in broad categories (e.g. Entertainment, News, etc.). The classification was perfomed automatically using ChatGPT-4o, and all domains were manually verified by one of the researchers (Appendix C, Table A8). After this classification, the comparisons were computed considering lists of website categories ([[News_1, Political_1, ...] vs. [Political_1, News_1...]), instead of single URL lists. As search results may contain multiple websites of the same category on a single results page, converting these lists into sets (i.e., removing duplicates) would distort the comparison and produce unfair *RBO* values. Therefore, X_d in (1) was adapted to use a set-based overlap for lists with repeated elements and defined as the count of minimum occurrences of each element present in both lists up to that depth:

$$X_d = \sum_{x \in S_d \cap T_d} \min\left(\mathrm{count}(x, S_d),\ \mathrm{count}(x, T_d)\right). \tag{3}$$

where $\mathrm{count}(x, S_d)$ is the number of times element x appears in the top d results of list S. For example, consider the top-5 elements of two lists: $S_5 = [a, b, a, c, d]$, $T_5 = [a, a, b, e, a]$. A regular set-based overlap calculation would score 2 (for a and b). However, since a appears at least twice in both lists, it should contribute more to the overlap. Equation 3, by considering the minimum number of occurrences of each element in both lists, correctly accounts for repeated elements and provides a more accurate measure of similarity.

Moreover, as Hebrew is much less spoken than the other languages and the availability of local sources per country could influence the results, we also compared the overall number of unique URLs, domains, and news shown to the bots by location. We found no concerning disparities – if anything, bots located in Israel and Saudi Arabia were often shown a greater diversity of results (see Appendix A, Figures A6, A7 and A8).

3) QUANTIFYING DIFFERENCES IN LEANING

To quantify leaning, the URLs from domains classified as "News" were selected from the 8 queries that were common to all three bot types (see Appendix A, Table A1). To minimize coder bias, these articles were scraped, all domain-related information removed, and their title and content collected and automatically translated to English, using the Google Translate package for Python, which provided an interface to Google's translation API.

Two approaches were used to quantify leaning as pro-Israel, slightly pro-Israel, neutral, slightly pro-Palestine or pro-Palestine: a) prompting the GPT-4o API (Appendix D, Figure A17), and b) tasking workers hired through the MTurk platform (Appendix D, Figure A18). In both cases, the prompt read: "Pro-Israel or pro-Palestine means that the text favors the narrative of that side of the conflict, or elicits sympathy towards it." MTurk workers who participated in text classification were subject to two attention controls: one during the recruitment stage, and another during the classification task. Each article was classified by a minimum of two and a maximum of three MTurk workers. For this analysis, we retained only the articles where at least two independent workers attributed the same classification, which corresponds to 65% of the total dataset. Thus, our leaning analysis is based exclusively on articles with total inter-coder agreement.

IV. RESULTS

We organized the results as answers to the main questions guiding our work. We ask if our bots (or proxy-users) see different results when querying search engines and, if so, how user features affect those differences (section IV-A). This was done incrementally, by varying the level of bot customization: first varying only the location (*Type 1*), then varying both location and browser language (*Type 2*), and finally varying location, browser language, and browsing history (*Type 3*). Next, we tested whether disparities in result URLs also correspond to meaningful differences in content (section IV-B) and/or content leaning (section IV-C) towards Israel or Palestine.

A. QUANTIFYING THE IMPACT OF PROFILES AND OF QUERIES ON SEARCH ENGINE RESULTS

1) DOES GEOGRAPHICAL LOCATION INFLUENCE SEARCH ENGINE RESULTS?

As discussed in the methods and summarized in Table 1, search engines typically report customizing results based on location or proxy location, and this may prove problematic

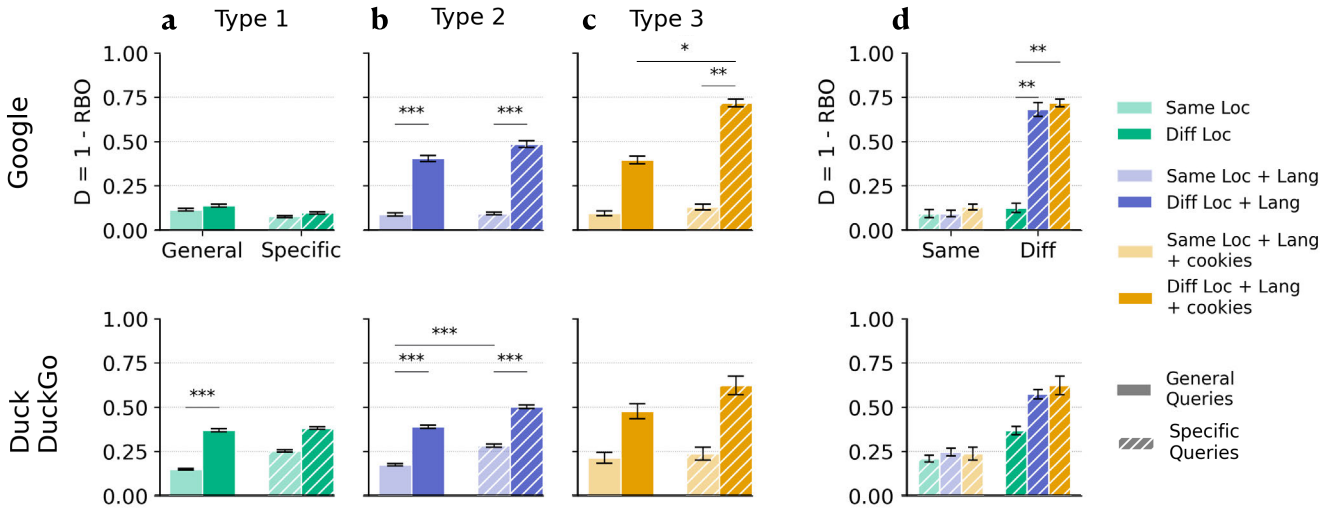


FIGURE 2. URL Dissimilarity Results ($D = 1 - RBO$, where *RBO* is a similarity metric). Higher values of *D* indicate that the lists of search results differ more substantially, especially at the top of the page, whereas smaller values of *D* indicate that differences are infrequent and occur mainly at the bottom of the results page. (a) Average *D* values between search engine results for *Type 1* bot pairs in the same location (light green) and different locations (dark green), for general (left) and conflict-specific (right, dashed) queries. (b) Average *D* for *Type 2* bots with the same (light violet) and different (dark violet) locations and browser languages, across query types and search engines. (c) Average *D* for *Type 3* bots with the same (light yellow) and different (dark yellow) location, browser language, and browsing history (cookies). (d) Average *D* for the 8 common conflict-specific queries across bot types, comparing bots with the same specifications (right) and different (left) across types (colors). Error bars represent 95% confidence intervals based on the bootstrapped samples from RBO values for the reported category comparisons (n=10,000 and seed=42). P-values were calculated with Mann-Whitney U test and adjusted with Bonferroni correction.*** denotes significant differences with p-value less than 0.001, ** less than 0.01, and * less than 0.05. Statistical details in Tables A2 and A3.

when querying on geopolitical conflicts. To examine how location affects search engine results, we deployed *Type 1* bots in four distinct regions: Israel (IL), Saudi Arabia (SA), United States (US), and Brazil (BR). These were directed to query both Google and DuckDuckGo about the the conflict (conflict-specific queries) or neutral topics (general queries). The first page of the results, with their respective ranks, was collected and the dissimilarity metric D (2) computed.

Figure 2, panel **a)** (green), summarizes these results, by comparing dissimilarity values (2) for different query types (general on the left and conflict-specific queries, dashed, on the right) for each search engine (vertically). Light green corresponds to the average *D* values for bots deployed at the same location (e.g. one BR bot compared to another BR bot) and in dark green the bots were deployed at different locations (e.g. BR bots vs US bots), with a high *D* meaning that (the top) results were more dissimilar.

We found a tendency of larger variation between locations (dark green) than within locations (light green), with the difference more pronounced for DuckDuckGo (p-values in Appendix B, Table A2) than for Google. This contradicts previous findings [14], [17], as in our results different locations do not seem to play a significant role.

Finding 1: Unlike anticipated (from previous research and from Google's privacy policy), location alone does not significantly impact Google's search results. For DuckDuckGo, it does play a role (in line with their stated policy), and this effect is stronger for lower ranking results.

2) DOES BROWSER LANGUAGE IMPACT SEARCH ENGINE RESULTS?

In the second phase of our experiment, we configured *Type 2* bots not only to be deployed from different locations, but also to interact with the search engines using the primary language of each location (see Methods, section III-A). These bots performed the same general and conflict-specific queries as before, under the hypothesis that this combination would further intensify the customization of search outcomes.

Figure 2, panel **b)** (violet), shows this: changing browser language to the local language leads to a greater divergence in search results for both search engines. These differences are larger when comparing bots from different locations (dark violet bars), and for conflict-specific queries (dashed bars), regardless of query length (Appendix B, Figure A11). In particular, Google search results become very diverse, but both engines show an average difference of 4 out of 10 results, with at least one different result in the Top 3 (Appendix B, Figure A9).

Finding 2: The combination of location and browser language increases the personalization of search results, especially for Google. This effect is more prominent for conflict-specific queries.

3) DOES BROWSING HISTORY IMPACT SEARCH ENGINE RESULTS?

It is debatable the extent to which search engines personalize the displayed content based on the user's past browsing experience. DuckDuckGo claims not to collect or use

personal data, presumably including browser history, [12], and Google claims not to personalize results unless users are logged-in to their Google accounts [44] (Background, Table 1). To test this, we directed the bots to visit (and collect cookies from) 20 general websites or 20 websites related to the conflict prior to querying the search engines (*Type 3* bots). If search engines do not personalize results based on browsing history, these cookies should have no impact, and the results would mirror those of *Type 2* (location and language) bots. However, Figure 2, panel **c)** (yellow) shows that the average *D* is generally even higher across both platforms, particularly for different locations (dark yellow bars) and for conflict-related queries (dashed bars), despite the users (bots) having no Google (or other) accounts.

This trend remained consistent when we compared the average *D* values obtained using the 8 queries common to all 3 bot types – details in the Methods section III-B – as summarized in Figure 2-**d**. *Type 3* (yellow) bots retrieve the most variable results, and even more so for bots deployed with different locations, languages, and browsing history (darker colors on the right versus lighter colors on the left).

Since the results shown for *Type 2* and *Type 3* were collected a few days apart (Appendix B, Figure A1), we also tested whether the observed increase in *D* could be attributed to fluctuations in browser language customization rather than the presence of cookies. As *Type 3* bots were always deployed alongside control bots with no browsing history (i.e., stateless bots equivalent to *Type 2*), we explicitly analyzed whether, when controlling for location and language, comparing the results of simultaneously deployed bots with and without browsing history lead to different *D* values. We found that, on average, bots with a conflict-related browsing history were twice as likely as stateless bots to receive a different top-3 result (see Figure A12 in Appendix B). This suggests that, even within the same location – where baseline result variability is low – the presence of cookies increases divergence in results.

Finding 3: Despite search engine claims of limited or no personalization based on user information, browsing history influences search engine results, particularly for conflict-specific queries on Google.

B. DIFFERENCES IN SEARCH ENGINE SUGGESTED CONTENT

One simple explanation for some of these differences, especially between *Type 1* and *Type 2* bots, could be that the search engines are directing the bots to very similar content but in different local domains. To investigate this, we repeated the analysis from Figure 2, this time comparing domains (e.g. wikipedia.org, newyorktimes.com) instead of URLs. This accounts for cases like `wikipedia/en/palestine` vs. `wikipedia/pt/palestina`, which are distinct URLs but may offer comparable content in different languages. However, the results closely mirrored those of Figure 2. This implies that: 1) different language versions of the same page were not a major contributor to the observed differences, and 2) bots with varying locations and features are shown different domains – not just different pages within the same domain (Appendix C, Figure A13).

Moreover, even when comparing domains, differences remained more pronounced for conflict-specific queries than for general ones, despite the ''general'' queries covering a wide range of topics. As the observed effect may arise from greater local diversity of content – such as an increased presence of local news articles on the conflict – we further classified all identified domains by category (e.g., News, E-Commerce, Sports; see Methods section III-C3, Figure 3-**b**, Appendix C - Table A8, and Appendix C - Figure A16) and repeated the analysis using these broad groupings.

Since the number of categories is much smaller than that of URLs or domains, the variability measured by *D* is expected to be lower. Nonetheless, Figure 3-**a** shows that *D* remains higher when comparing bots with different characteristics (light vs. darker colors), with particularly elevated values for *Type 2* and *Type 3* bots, especially on Google (top row) and for conflict-specific queries (dashed bars). This is somewhat surprising, as bots issuing general queries typically encounter more result categories per page. Notably, this variability persists even when a single category, ''News'', dominates the result set (Figure 3-b).

Together, these three analyses – URLs, domains and type of content – strongly support the finding that both search engines, but particularly Google, personalize the results and content of political and sensitive queries to a greater extent, than for general and more popular queries.

Next, we examined how the type of suggested content varied. As bots are personalized via browser language (*Type 2*) and browsing history (*Type 3*), those performing general queries (Figure 3-**b**, left panel) tend to be shown similar content categories. In contrast, for conflict-specific queries (right panel), bots are increasingly shown ''News'' content (light blue), particularly on DuckDuckGo (bottom row). For conflict-specific queries on Google, we also observed that the number of unique news domains grows with bots' profiling, which corresponded to an increase of unique domains being shown only to bots of a certain location (Figure 3-**c** and Figure A14 in Appendix C).

This reinforces the argument that search engines are leveraging bot information, including browsing history, to tailor results and rankings: bots previously exposed to ''News'' (*Type 3* bots) have a higher chance of encountering a ''News'' result, per page, for both search engines, with these results also being more country-specific, even for identical queries.

Finding 4: Distinct bot profiles are directed not just to local versions of the same URLs, but to fundamentally different types of results. These disparities are particularly pronounced for conflict-specific queries rather than general unpolitical ones, especially for bots that have collected cookies.

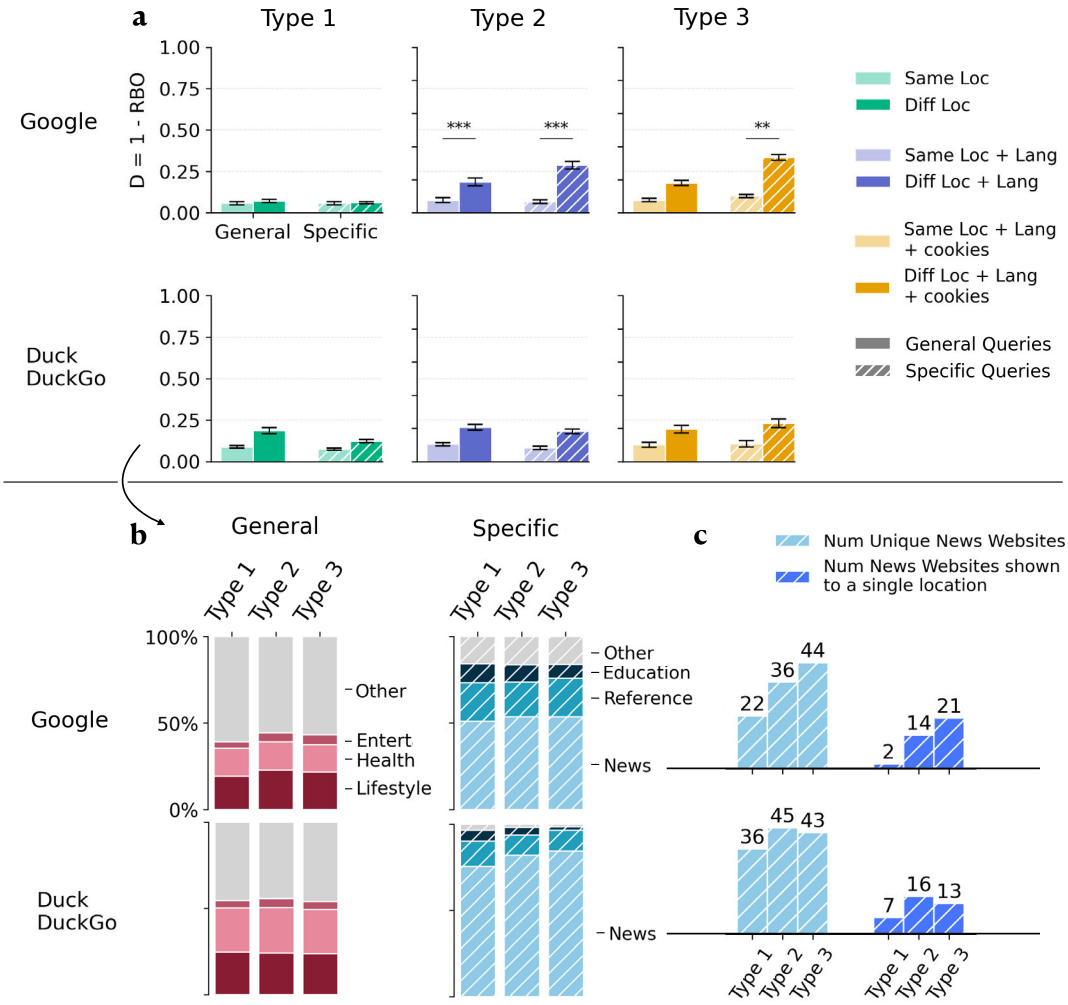


FIGURE 3. Content Dissimilarity Results ($D = 1 - RBO$). Higher values of D indicate that the lists of search results differ more substantially, especially at the top of the page, whereas smaller values of D indicate that differences are infrequent and occur mainly at the bottom of the results page. (a) Average value of $D = 1 - RBO$ for website categories of results shown to bots with the same (lighter colors) or different features (darker colors). Left: general queries; Right and dashed: specific queries. Error bars represent 95% confidence intervals based on the bootstrapped samples from RBO values for the reported category comparisons (n=10,000 and seed=42). P-values were computed using the Mann-Whitney U test and adjusted via Bonferroni correction. Significance is indicated as follows: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$ - more information in Appendix C, Table A9. (b) Top three most frequent categories of websites in results for general (left) and conflict-specific (right) queries, for the common queries (8) performed by all bot types. (c) Number of unique news websites for the specific query results for the three bot types.

C. LEANING OF SEARCH ENGINE RESULTS

Even if the results and the page contents differ, and even if these differences increase with bot profilling, it could still be argued that the variance stems from algorithmic randomness and is ultimately inconsequential. Conversely, if search engine algorithms use information such as location and browsing history to customize geopolitical content, this could reinforce pre-existing views (for a detailed and real example, consult Appendix Table A5).

To explore this, we examined whether content differences also reflect differences in political leaning. We focused on the ''News'' category of the conflict-related searches for two reasons: this topic dominates the query results (Figure 3-**b**), and news articles strongly influence opinion formation [73]. This analysis was done in two steps: we successfully retrieved the content from over 90% of the 714 unique news URLs, and then classified each as pro-Israel, pro-Palestine, or neutral, using a combined approach of human and automatic classification (details in Methods). For the human coding, the analysis was done on the 465 articles that received exactly the same classification by at least two independent MTurk workers.

We highlight five important observations. First, ChatGPT (Figure 4-**a**) labeled significantly more content as neutral (gray) than human coders (Figure 4-**b**). Second, MTurk classifiers mark more content as pro-Israel (shades of turquoise) than pro-Palestine (shades of orange). Third, MTurk workers disagreed more often with ChatGPT's pro-Palestine

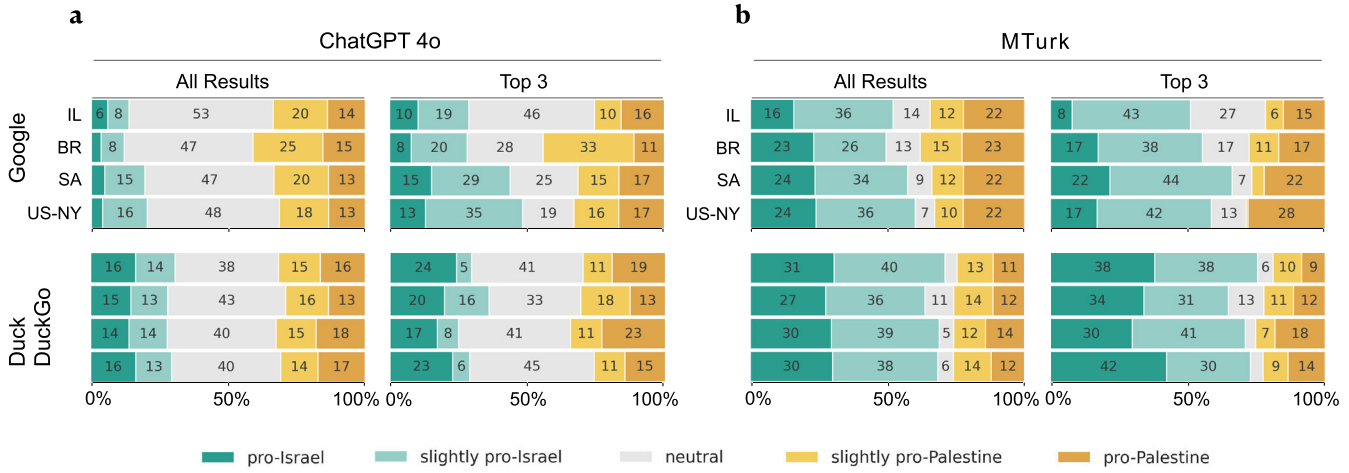


FIGURE 4. Percentage of news articles classified as being "neutral" (grey), "pro-Israel" and "slightly pro-Israel" (turquoise shades) and as being "pro-Palestine" and "slightly pro-Palestine" (orange shades), for each country (y-axis) and search engine (top, Google, bottom, DuckDuckGo). (a): ChatGPT 4o classification and (b): MTurk classification.

classifications, than with pro-Israel ones (Appendix D, Figure A19 and Figure A20). In fact, the calculated Cohen's Kappa value [74] and the Krippendorff's α [75] were 0.28 and 0.23 respectively, indicating a fair, but low agreement that exposes the task's complexity and subjectivity – more details on classification agreement in Appendix Figure A19 – and per search engine and location in Figure A20. Fourth, leaning shifted based on ranking: while more articles overall were labeled as pro-Palestine (29%) than pro-Israel (20%), the top three results – especially on Google – favored pro-Israel content (Figure 4-**a**), for both ChatGPT and MTurk (Figure 4-**a**, last column).

Finally, we analyzed how domain-level leanings influenced results. We asked whether some sites consistently leaned one way (expected), and whether search engines favored such domains in personalized results (less expected). We found that some domains consistently published content classified as predominantly leaning towards one side of the conflict (e.g., Al-Monitor, The Arab Weekly, Jewish Telegraphic Agency; see Appendix D, Table A10). Notably, these domains appeared more often on Google for the *Type 3* bots, with both classifiers indicating fewer neutral results (Appendix C, Figure A14).

Finding 5: Bots at different locations receive content with different leanings on the conflict, according to both ChatGPT and MTurkers, despite ChatGPT classifying as neutral news that humans perceive as biased.

V. DISCUSSION AND LIMITATIONS

Here, we discuss some of the study's weaknesses, our attempts at minimizing possible threats to validity, and possible impacts of our findings:

1. Non-human queries: Despite our best efforts, the bots do not perfectly mimic humans and, if identified as bots, the search engines might handle their queries differently than they would for human-generated traffic. To minimize this, we coded the bots to act more human-like (introducing delays, natural "typing", avoiding VPN farms, etc.) and we have reasons to believe that they were only rarely identified as such (e.g. we received few "captcha" requests and only some queries failed, with Google trials succeeding about 72% of the time and DuckDuckGo over 98% of the time). Also, the bots have none (*Type 1* and *Type 2*) to limited (*Type 3*) browsing history, and this could limit personalization of the results. However, that even such a conservative approach managed to reveal significant differences for conflict-specific queries should raise important alarms, as these are likely to be even higher for human queries. In the future, audits could enhance the bot's realism by increasing browsing history (breadth and size) and adding features such as search language or social media profiles.

2. Combinatorial complexity: The incremental and combinatorial design of our approach led to a rapid increase in the number of variables, making exhaustive testing complex and expensive. *Type 3* bots only queried approximately one third of the original list and we used a conservative approach for their selection: the selected 8 queries were the ones for which *Type 1* and *Type 2* bots had the smallest differences in *D* between locations. Thus, it is possible that if other queries were chosen the observed effects would have been larger, but it is unlikely they would have been smaller.

3. Geographical IP limitations: In the absence of IP addresses for Palestine or for countries historically very supportive of Palestine (such as Iran or Lebanon), the choice of Saudi Arabia may have reduced the bot exposure to pro-Palestine content – despite having a majority Sunni population and an official pro-Palestine stance. This omission, together with the English language queries, could underestimate the pro-Palestine perspectives in the overall dataset, leading to reported smaller bias / gap than the possible real

one. Moreover, despite using residential proxies, which are typically less subject to controls, the bots occasionally failed, for unknown reasons (see Appendix A, Figure A5).

4. Dynamic nature of search engines: That search engine algorithms are both fast-evolving and proprietary means that even minor variations in the timing of our audits could yield different results. To control for such temporal fluctuations, particularly during such a rapidly unfolding situation, we repeated the same queries for stateless *Type 2* bots three times and observed no temporal trend (Appendix E, Figure A22). Future studies should perform audits more frequently and for longer periods, and include more queries, search engines, and Large Language Model (LLM)-based interfaces.

5. Analysis of leaning: There is no objective metric of leaning and there were significant differences in classification between ChatGPT and human coders. In particular, news that humans identified as more favorable to Israel were likely to be considered neutral by ChatGPT. This is, in itself, a highly interesting result that deserves future research: why does this happen? Are human coders better at identifying subtle bias, or is ChatGPT more impartial? Could such apparent impartiality be a consequence of its training, designed to promote neutrality, particularly around political issues? Future work should have more coders, from different locations, including expert coders, and explore classifications around different topics. In addition, it can be argued that the identified bias in news articles, in favor or against a particular perspective on the conflict, is the responsibility of news outlets and not of the search engines. However, it is ultimately the algorithmic decision of the search engines to prioritize specific content (whether it is driven by search engine optimization techniques, higher engagement, and/or information favoring a particular side of the conflict). That biased content might be shown has significant consequences: just as social media algorithms surface the most popular content – often aligned with users' viewpoints – search engine curation can also influence the diversity and polarization of the information [28], [76]. It might be even more impactful as people tend to place a higher trust in search engine results, as discussed in section II.

VI. CONCLUSION AND FUTURE WORK

This paper presents a novel methodology for auditing search engine results using web crawlers designed to mimic different ''human-like'' online profiles. By focusing on the Israeli-Palestinian conflict and examining how factors such as location, browser language, and browsing history affect search results across Google and DuckDuckGo, we demonstrated that there are significant differences in search engine results,as these features vary.

We found that location alone had limited impact on Google search results but led to noticeable variation in DuckDuckGo. However, when combined with other identity and regional signals, specifically browser language and browsing history, it lead to significant increases in result variability, particularly for queriesrelated to the conflict. Importantly, this effect is not limited to differences in URLs, but extends to the types of content displayed, such as news or political content. Contrary to some earlier studies [18], [20], [26], our results revealed that browsing history does induce notable differences, particularly in the case of Google, and this may further amplify the effects of information bubbles or echo chambers.

Several reasons can explain these differences. First, we use a different approach, relying on systematic bot tailoring, instead of specific user profiles. Real users are likely to be profiled in highly non-linear ways, and this complexity might mask recommendation differences. Second, we are comparing across continents, while previous studies were mostly US based. In fact, similar patterns were reported by Paramita et al. [77], who found that Covid-19 image search results varied substantially across European regions, and that queries in different languages often produced completely non-overlapping results. Taken together, this suggests that the targeting of local results imposes strong filters on the information users can access. Third, these algorithms are fast evolving, and it is possible that our analysis is uncovering current personalization trends – and that the previous findings are no longer applicable. If so, this means that search engines may no longer fully adhere to their stated personalization policies: browsing history produced even more divergent results, despite their claims and our use of a controlled and relatively modest history. This is worrying and consistent with published research on Google News [24].

Another key finding is that customization of search results was more pronounced for queries related to the Israeli-Palestinian conflict than for general ones, even when some of which deliberately selected for cultural sensitivity and potential geographic variation, such as ''recipe'' and ''best books 2024''. This is concerning: while everyday searches showed limited personalization, politically sensitive queries were strongly shaped by user profiles. In other words, the algorithmic personalization that subtly adjusts recipe or book suggestions appears amplified in a context where divergent content may reinforce biases or deepen polarization. Importantly, this work demonstrates that while tailoring search results based on region is often considered ''neutral'' or benign, this assumption fails in the context of geopolitical conflicts. When individuals across political borders are shown different content – potentially with opposing political leanings – the consequences can be serious, unpredictable, and difficult to measure.

Future work should focus on the impacts of such personalized results. A similar approach can be applied to other sensitive situations, such as pre-electoral periods (e.g., how do search engines reply to broad and neutral electoral queries across countries or states?), harmful profiling (e.g., are users with a history of visiting low credibility websites more likely to be shown misinformation?), or even bias amplification (e.g., are users profiled as male recommended sexist content?). Such systematic analysis can also help

disentangle how specific components of browsing history (e.g., cookies, session data) contribute to personalization outcomes; however, the way forward in the online tracking space is uncertain. Not only is there strong resistance from the platforms to sharing fundamental data and metadata, but even the relative importance of such data keeps changing. In 2022, Google led a shift away from third-party cookies, then seemingly turned around [78] and, more recently, announced a greater integration with LLM-based results [79], further blurring the line between traditional search results and generative AI responses. Importantly, our results suggest that content judged as partial by human coders may be interpreted as impartial by LLMs such as ChatGPT. As these models are increasingly used, such apparent neutrality may contribute to amplifying trust in search engines, while masking troubling biases.

Overall, our findings underscore the urgent need for greater transparency and auditing of these rapidly evolving platforms, used by millions daily, often for sensitive topics such as the one studied here.

In the European Union, the Digital Services Act specifically states that very large online search engines must provide access to data and/or code deemed necessary to monitor and guarantee compliance on areas such as privacy protection and personalization. This regulation also includes a mechanism for researcher access and academic auditing around systemic threats. Using such a mechanism, it should be possible to extend this methodology or request access to relevant information on algorithmic recommendations, crucial to our understanding of what is shown to which user and why. This includes providing sufficient researcher support, standardizing and streamlining data access, sharing metadata and indexes, and allowing controlled A/B testing. To mitigate search engine bias and opaque targeting, platforms should adopt strategies such as diversification prompts (e.g., creating a tab to ''explore different perspectives''), disclosure of ranking and personalization mechanisms, and access to clear documentation. In summary, collaborative governance mechanisms, between platforms, researchers, and regulators, are essential to ensure effective oversight and uphold the public interest, increasing transparency and algorithmic explainability of the entire online search environment.

Building on these directions, future work should further adapt this methodology to study AI-generated responses, which are becoming increasingly pervasive in both search results and used for everyday tasks.

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